

Growth Incrementality & Targeting Lab

Decision: test uplift against propensity at equal reach.

Advance both rankings to a fresh randomized policy comparison. This benchmark does not establish that uplift beats conversion propensity, and it does not justify production rollout on its own.

Evidence

A seeded 10% sample of Criteo v2.1 contains **1,399,703 users**, with **279,824** in a final holdout. Feature groups stay within one split. The linear T-learner won validation against two boosted candidates.

Assignment to advertising increased final-holdout conversion by **0.105 percentage points** (95% CI 0.057 to 0.152). At 20% reach, extra conversions per 1,000 eligible users were:

Policy	Estimate	95% interval
Selected uplift	1.00	0.56 to 1.45
Conversion propensity	1.06	0.60 to 1.51
Random targeting	0.21	0.11 to 0.30

Paired uplift minus propensity: **-0.055** (95% CI -0.199 to 0.088); superiority is not established. Uplift minus random: **0.792** (0.435 to 1.149).

Budget implication - hypothetical only

At 100,000 eligible users, 20% reach, \$0.20 per treated user and \$100 per extra conversion: **100.1** estimated extra conversions, **\$4,000** spend, and **\$6,009** net incremental value (95% sampling interval \$1,566 to \$10,453). These are scenarios, not observed revenue or profit.

Boundaries and next step

Privacy subsampling changes incrementality. Anonymized features, pooled tests, missing experiment IDs and only 80 control conversions limit interpretation. Pointwise intervals exclude model-selection and transport uncertainty. Prespecify the audience, conversion window, economics and meaningful policy difference; then randomize equal-reach policies with contamination and margin guardrails.

Source: Criteo AI Lab; Diemert et al. (2018), A Large Scale Benchmark for Uplift Modeling. Data-derived report: CC BY-NC-SA 4.0. Independent work; no Criteo endorsement. Methods and code: github.com/IamGRootMSE/growth-incrementality-lab